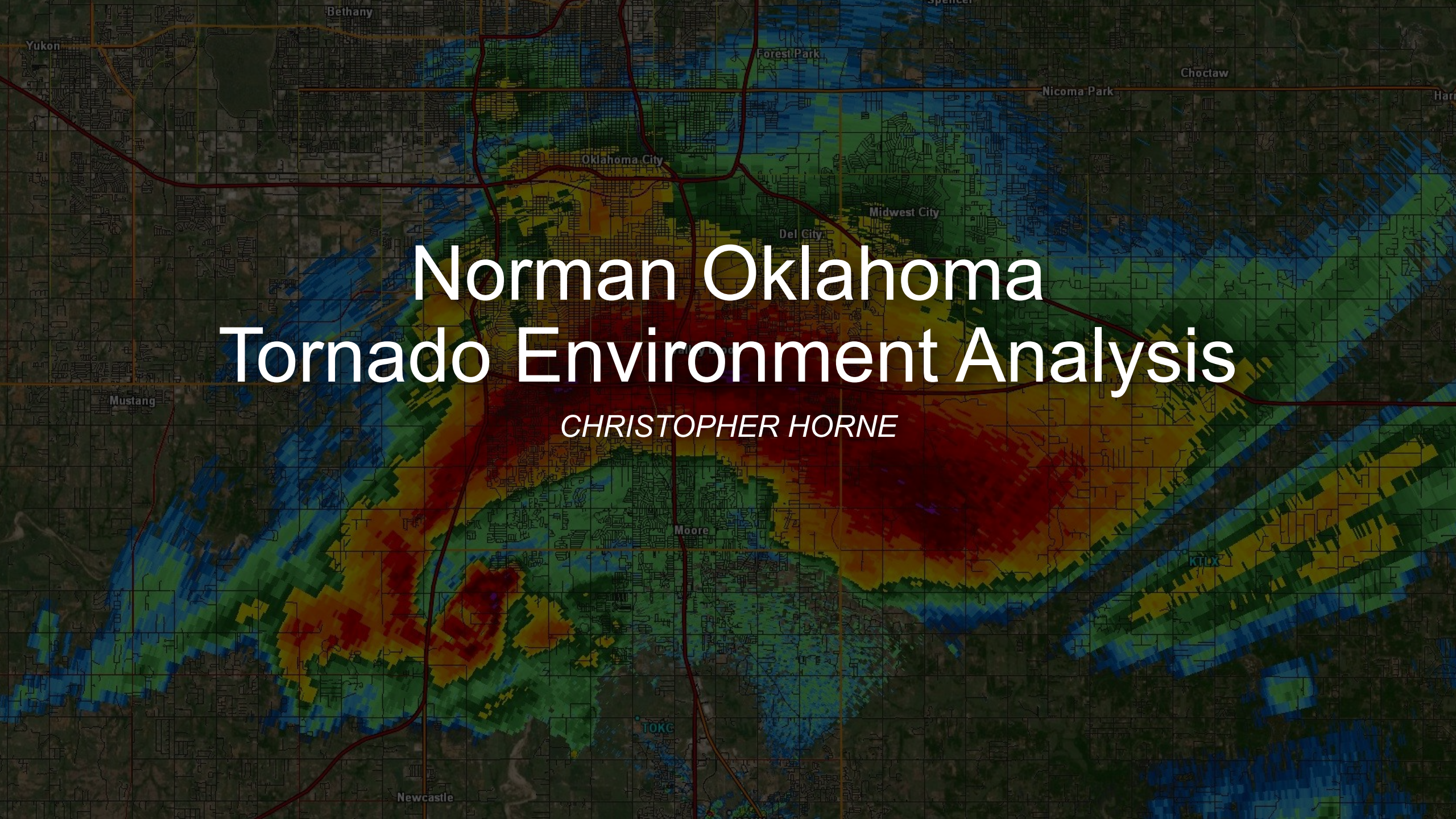
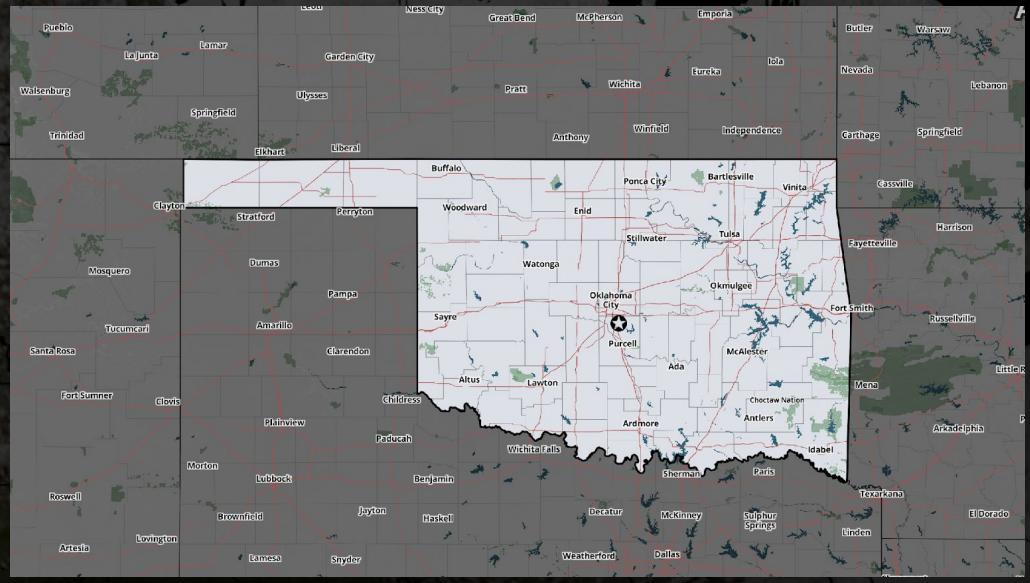


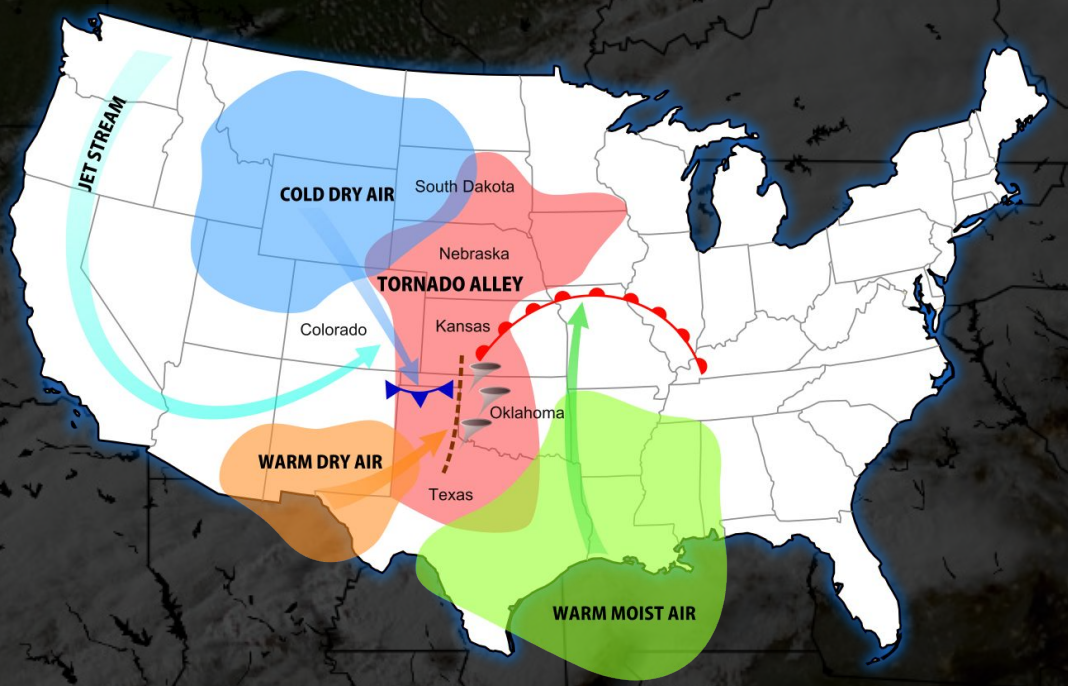
Norman Oklahoma Tornado Environment Analysis

CHRISTOPHER HORNE

INTRODUCTION

- ★ Study area: Norman / OKC Region
- ★ Located in the Southern Plains tornado corridor
- ★ Inland, mostly flat terrain with little coastal influence
- ★ Gulf moisture commonly moves north into Oklahoma
- ★ Dry air often advances east from the High Plains
- ★ Drylines, fronts, and surface lows often trigger storms
- ★ Low-level jet increases shear and storm-relative helicity
- ★ Elevated mixed layer creates steep lapse rates and CAPE
- ★ This project compares tornado environments across EF ratings and different storm modes
- ★ Goal: examine how atmospheric ingredients relate to tornado destructiveness in central Oklahoma





SELECTED EVENTS

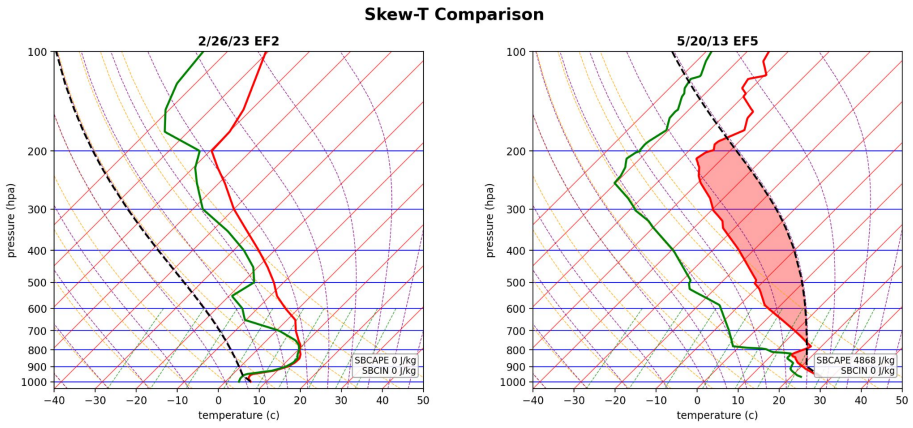
- ★ Bridge Creek Moore F5
  - May 3rd, 1999 → Incredible Violent Tornado Highest Wind Ever Record
- ★ Moore F3
  - May 8th, 2003 → Significant Tornado Mid Strength
- ★ Lake Thunderbird EF4
  - May 10th, 2010 → Strong Composite Environment
- ★ Norman EF1
  - April 13th, 2012 → Heavy CIN and Lower CAPE
- ★ Newcastle EF5
  - May 20th, 2013 → Extreme CAPE and Violent, Deadly Tornado
- ★ Norman EF1
  - May 6th, 2015 → High CAPE, Weak Tornado
- ★ Norman EF2
  - February 26th, 2023 → Occured in a QLCS Environment





SELECTED DATASETS

- ★ Soundings were accessed from the University of Wyoming Sounding Archive
- ★ Sounding station used: OUN / Norman, Oklahoma
- ★ Soundings provided vertical profiles of needed variables for calculation
- ★ For some events, OUN soundings were not fully representative of the tornado location or timing
- ★ In those cases, ERA5 reanalysis sounding data was used for a better environmental estimate



72357 OUN Norman Observations at 00Z 14 Apr 2012

PRES hPa	HGHT m	TEMP C	DWPT C	RELH %	MIXR g/kg	DRCT deg	SKNT knot	THTA K	THTE K	THTV K
1000.0	72									
969.0	345	19.0	17.3	90	12.99	70	3	294.8	332.2	297.1
950.0	516	20.6	19.2	92	14.97	128	12	298.1	341.7	300.8
939.8	610	20.2	18.8	92	14.79	160	17	298.6	341.8	301.2
925.0	747	19.6	18.3	92	14.52	165	25	299.3	341.9	301.9
911.0	879	19.0	17.3	90	13.83	173	31	300.0	340.7	302.5
907.2	914	18.7	17.1	90	13.72	175	33	300.1	340.5	302.6
875.4	1219	16.5	15.5	94	12.83	180	30	300.8	338.7	303.2
850.0	1471	14.6	14.2	97	12.12	180	31	301.4	337.3	303.6
844.7	1524	14.2	13.9	98	11.93	195	32	301.6	337.0	303.7
818.0	1795	12.2	12.2	100	11.03	191	30	302.2	335.1	304.2
814.7	1829	12.0	12.0	100	10.92	190	30	302.3	334.9	304.3
785.4	2134	10.1	10.1	100	10.00	190	28	303.5	333.6	305.4
757.3	2438	8.3	8.3	100	9.15	185	28	304.7	332.4	306.4
753.0	2486	8.0	8.0	100	9.02	187	28	304.9	332.2	306.5
729.8	2743	5.6	5.0	96	7.58	200	28	305.0	328.1	306.4
724.0	2809	5.0	4.3	95	7.24	201	31	305.0	327.1	306.4
714.0	2922	4.2	0.4	76	5.55	203	36	305.4	322.5	306.4
703.1	3048	5.3	-1.8	60	4.80	205	41	307.9	323.0	308.8
700.0	3084	5.6	-2.4	56	4.60	205	41	308.6	323.2	309.5
695.0	3143	6.2	-4.8	45	3.87	207	41	309.9	322.4	310.7
681.0	3309	5.6	-14.4	22	1.85	213	41	311.1	317.3	311.4
664.0	3515	4.2	-18.8	17	1.31	220	41	311.8	316.3	312.0
654.0	3638	3.2	-14.8	25	1.86	224	41	312.0	318.3	312.4
652.4	3658	3.0	-14.8	26	1.86	225	41	312.0	318.3	312.4
641.0	3800	1.8	-15.2	27	1.84	226	42	312.2	318.4	312.6
626.0	3990	0.6	-25.4	12	0.78	228	43	312.9	315.7	313.1
616.0	4119	-0.5	-27.5	11	0.65	229	44	313.1	315.5	313.2
605.0	4262	-1.5	-20.5	22	1.24	230	45	313.6	317.9	313.8
604.6	4267	-1.5	-20.5	22	1.25	230	45	313.6	317.9	313.8
559.2	4877	-7.5	-20.1	36	1.39	240	49	313.6	318.4	313.9
535.0	5223	-10.9	-19.9	47	1.48	238	51	313.6	318.6	313.9
520.0	5441	-12.7	-21.7	47	1.30	237	52	313.9	318.5	314.2
514.0	5530	-12.7	-17.7	66	1.86	236	53	315.0	321.3	315.4
502.0	5710	-12.9	-19.9	56	1.58	235	54	316.9	322.4	317.2
500.0	5740	-13.1	-20.1	56	1.56	235	54	317.0	322.4	317.3
496.6	5791	-13.4	-20.3	56	1.55	235	54	317.2	322.6	317.5
477.0	6096	-15.5	-21.2	62	1.49	245	50	318.4	323.6	318.6
451.0	6521	-18.3	-22.5	70	1.40	254	48	320.0	324.9	320.2
447.0	6588	-18.3	-22.7	68	1.39	256	47	320.8	325.7	321.0
443.0	6655	-18.5	-24.5	59	1.19	257	47	321.3	325.6	321.6
435.0	6792	-19.7	-32.7	31	0.57	260	46	321.5	323.6	321.6

METHODS

VARIOUS WIND SHEAR 0-1KM, 0-3KM, 0-6KM

$$Shear = \sqrt{\Delta u^2 + \Delta v^2}$$

- ★ Measures wind change with height
- ★ Shows storm organization potential
- ★ Strong 0-6km shear favors rotating supercells
- ★ 0-1km shear shows tornado spin

STORM RELATIVE HELICITY SRH 0-1KM, 0-3KM

$$c_{rx} = u_{mean} + 7.5\left(\frac{v_{shr}}{Shear}\right)$$

$$c_{ry} = v_{mean} - 7.5\left(\frac{u_{shr}}{Shear}\right)$$

$$SRH = \sum[(u_{top} - c_x)(v_{bottom} - c_y) - (u_{bottom} - c_x)(v_{top} - c_y)]$$

- ★ Measures low-level streamwise rotation
- ★ Important for mesocyclone strength
- ★ Higher SRH increases tornado potential
- ★ Calculated Bunkers needed for SRH calculation

LIFTED CONDENSATION LEVEL

$$T_L = \frac{1}{\frac{1}{T_d - 56} + \frac{\ln(T/T_d)}{800}} + 56$$

$$z_{LCL} = z_{sfc} + \frac{T - T_L}{\Gamma_d}$$

- ★ Estimates cloud-base height
- ★ Lower LCLs favor tornadogenesis
- ★ Used in the Fixed STP calculation

METHODS

MID-LEVEL LAPSE RATE 500-700MB

$$\Gamma = -\frac{\Delta T}{\Delta Z}$$

- ★ Measures cooling with height
- ★ Steeper lapse rates increase instability
- ★ Supports stronger thunderstorm updrafts
- ★ Also can be an indicator of strong to severe hail

SURFACE BASED CONVECTIVE AVAILABLE POTENTIAL ENERGY

$$SBCAPE = \int_{LFC}^{EL} g\left(\frac{T_{parcel} - T_{env}}{T_{env}}\right) dz$$

- ★ Measures buoyant energy for surface parcels
- ★ Higher CAPE supports stronger updrafts
- ★ Helps compare instability across events
- ★ Discrete supercells tend to produce more violent tornadoes because of open inflow

FIXED SIGNIFICANT TORNADO PARAMETER

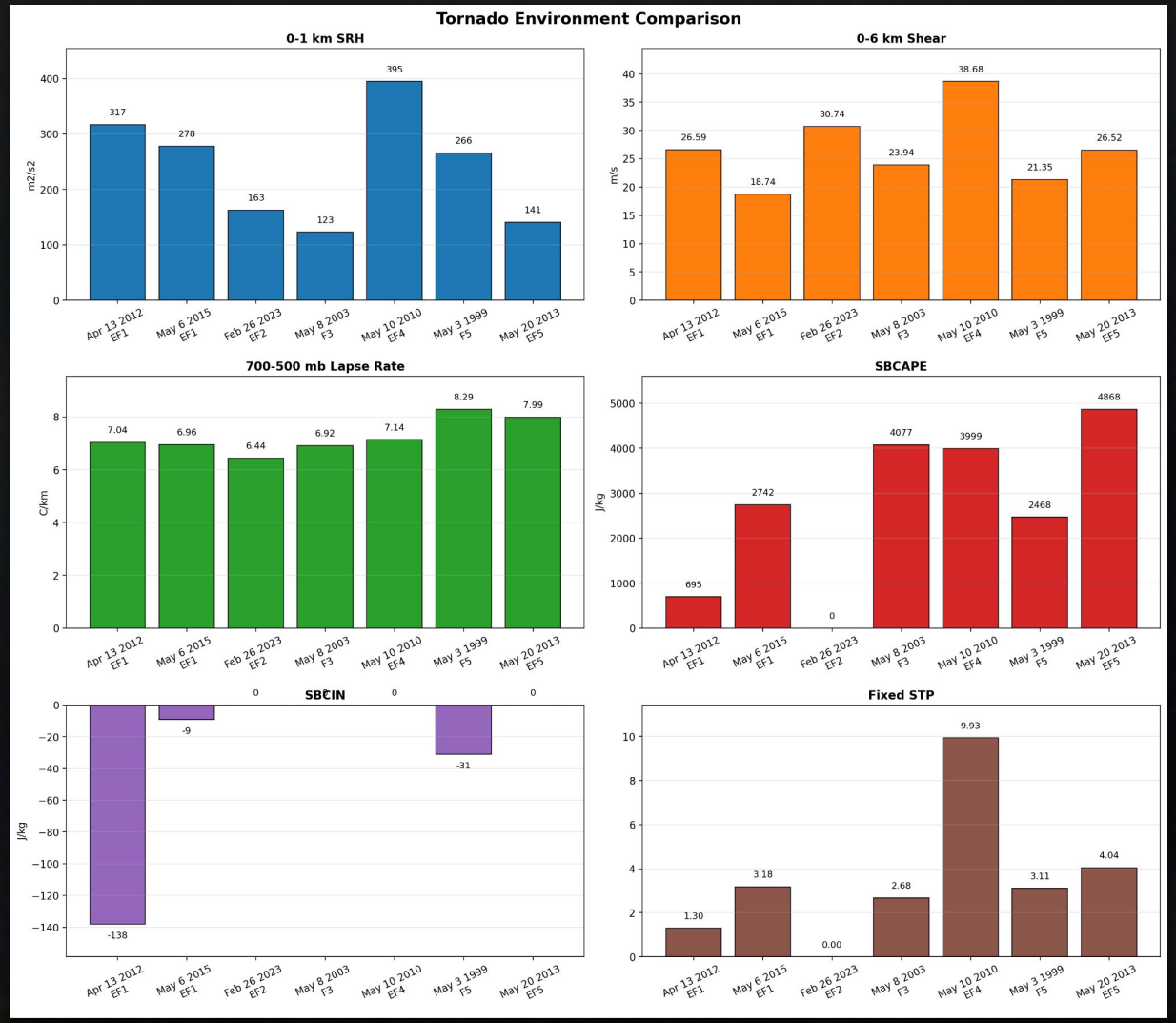
$$STP_{fixed} = \left(\frac{\text{SBCAPE}}{1500}\right) \times \left(\frac{2000 - \text{SBLCL}}{1000}\right) \times \left(\frac{\text{SRH01}}{150}\right) \times \left(\frac{\text{SHR06}}{20}\right)$$

- ★ Combines key tornado ingredients
- ★ Includes CAPE, shear, SRH, LCL, and CIN
- ★ Useful for comparing overall tornado favorability in a non QLCS environment

ANALYSIS

- ★ Compares each major tornado ingredient across all events
- ★ May 10, 2010 had the strongest SRH, shear, and STP signal
- ★ May 20, 2013 had the highest SBCAPE
- ★ May 3, 1999 had the steepest midlevel lapse rate
- ★ EF rating did not increase perfectly with any single parameter

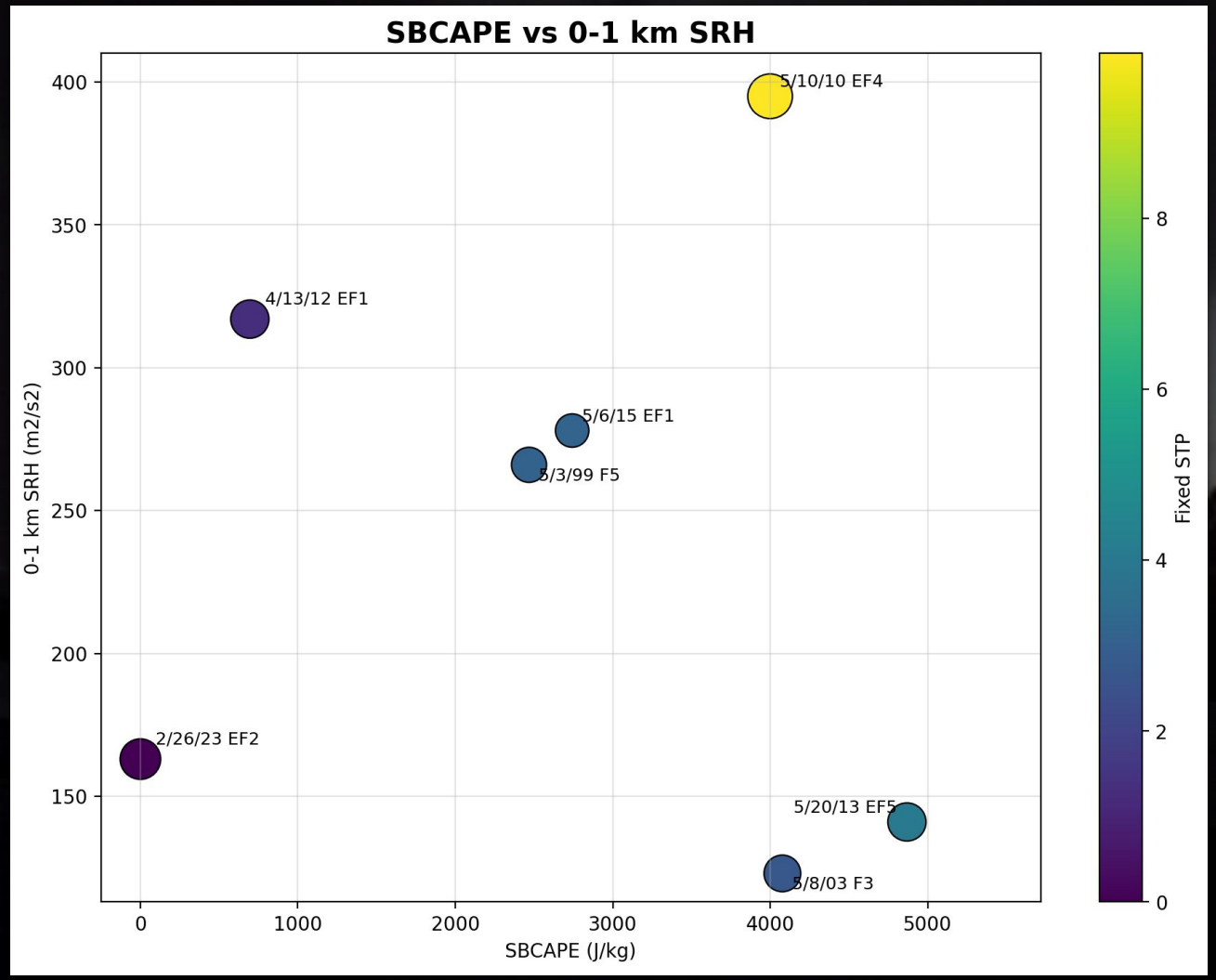
Tornado intensity depended on the overlap of ingredients rather than one individual value.



ANALYSIS

- ★ Compares instability against low-level rotation for each event
- ★ Strong tornado environments need both buoyancy and low-level shear
- ★ May 10, 2010 had the best overlap of CAPE and SRH
- ★ May 20, 2013 had extreme CAPE, but lower low-level SRH
- ★ Feb. 26, 2023 shows a QLCS tornado can occur with little SBCAPE

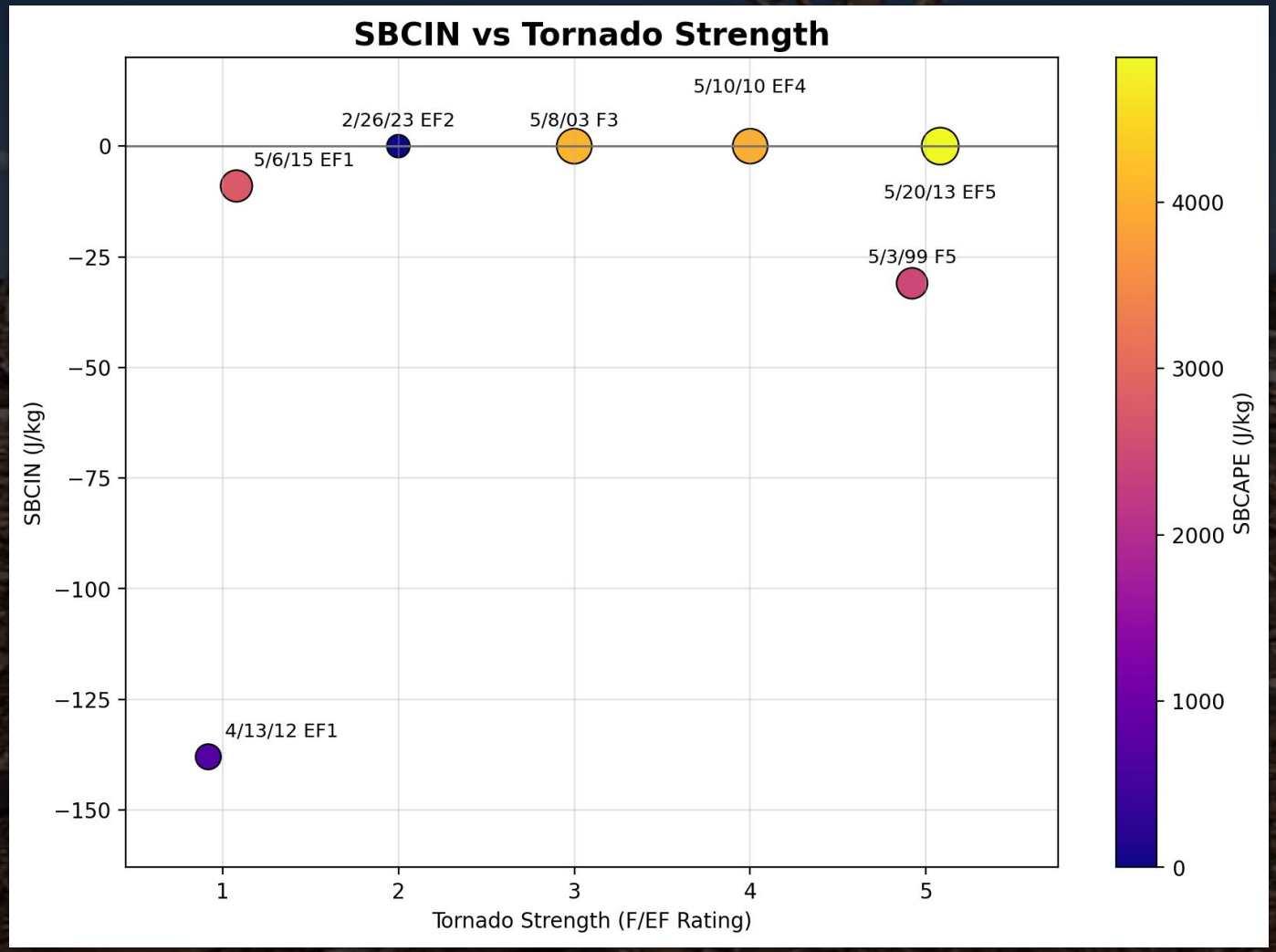
Tornado potential was strongest when instability and low-level rotation overlapped, not when only one ingredient was extreme.



ANALYSIS

- ★ Compares capping strength against tornado rating
- ★ CIN shows how much energy prevents surface parcels from rising
- ★ Stronger tornado cases generally had weak or near-zero CIN
- ★ Apr. 13, 2012 had strong inhibition, which likely limited storm potential
- ★ Low CIN allowed storms to access the unstable air more efficiently

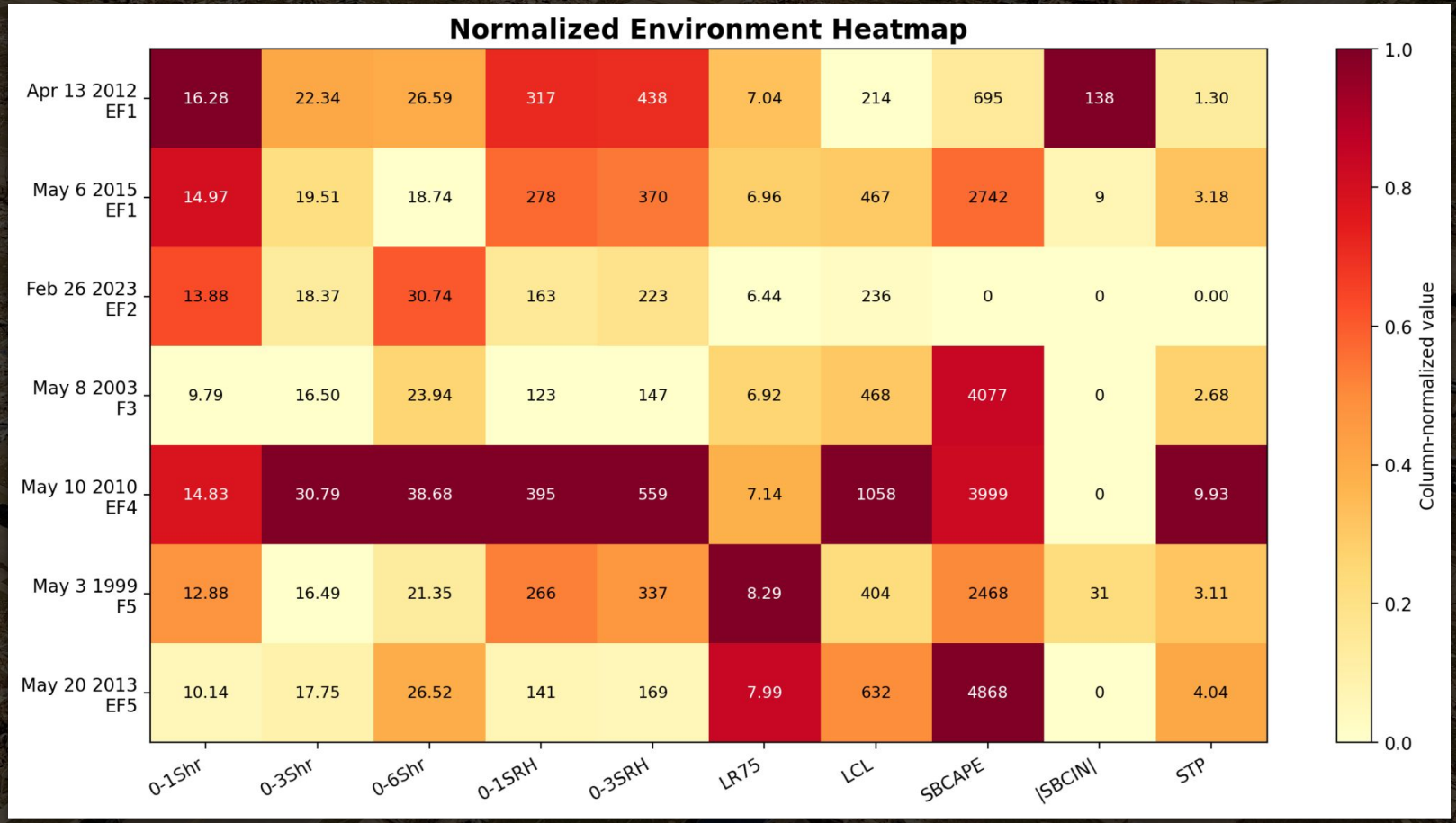
Stronger tornadoes were favored when inhibition was weak enough for storms to fully use the available instability.



ANALYSIS

- ★ Compares all parameters on the same relative scale
- ★ Darker colors show which ingredients were strongest for each event
- ★ May 10, 2010 had the strongest overall composite
- ★ EF5 cases were thermodynamically strong, especially CAPE and LRs
- ★ Feb. 26, 2023 QLCS stands out as more shear-driven than instability-driven

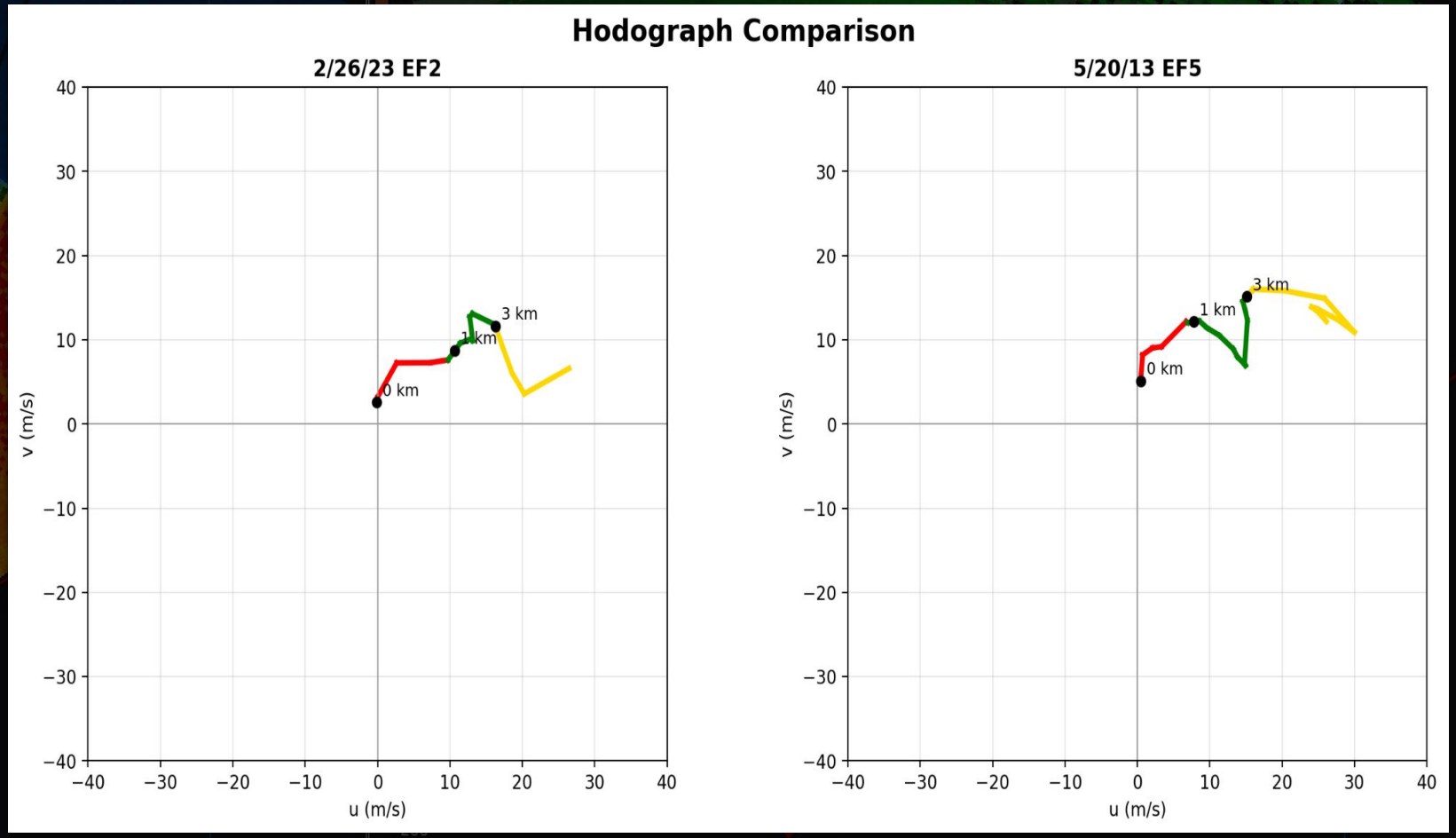
Each tornado event had a unique environmental fingerprint, even when the tornado ratings were similar.



ANALYSIS

- ★ Compares wind profiles between the QLCS EF2 and EF5 case
- ★ Hodographs show wind speed and direction changes with height
- ★ Curved hodographs indicate stronger turning and tornado potential
- ★ EF5 case showed a more classic supercell wind profile
- ★ EF2 QLCS case still had enough low-level shear for embedded rotation

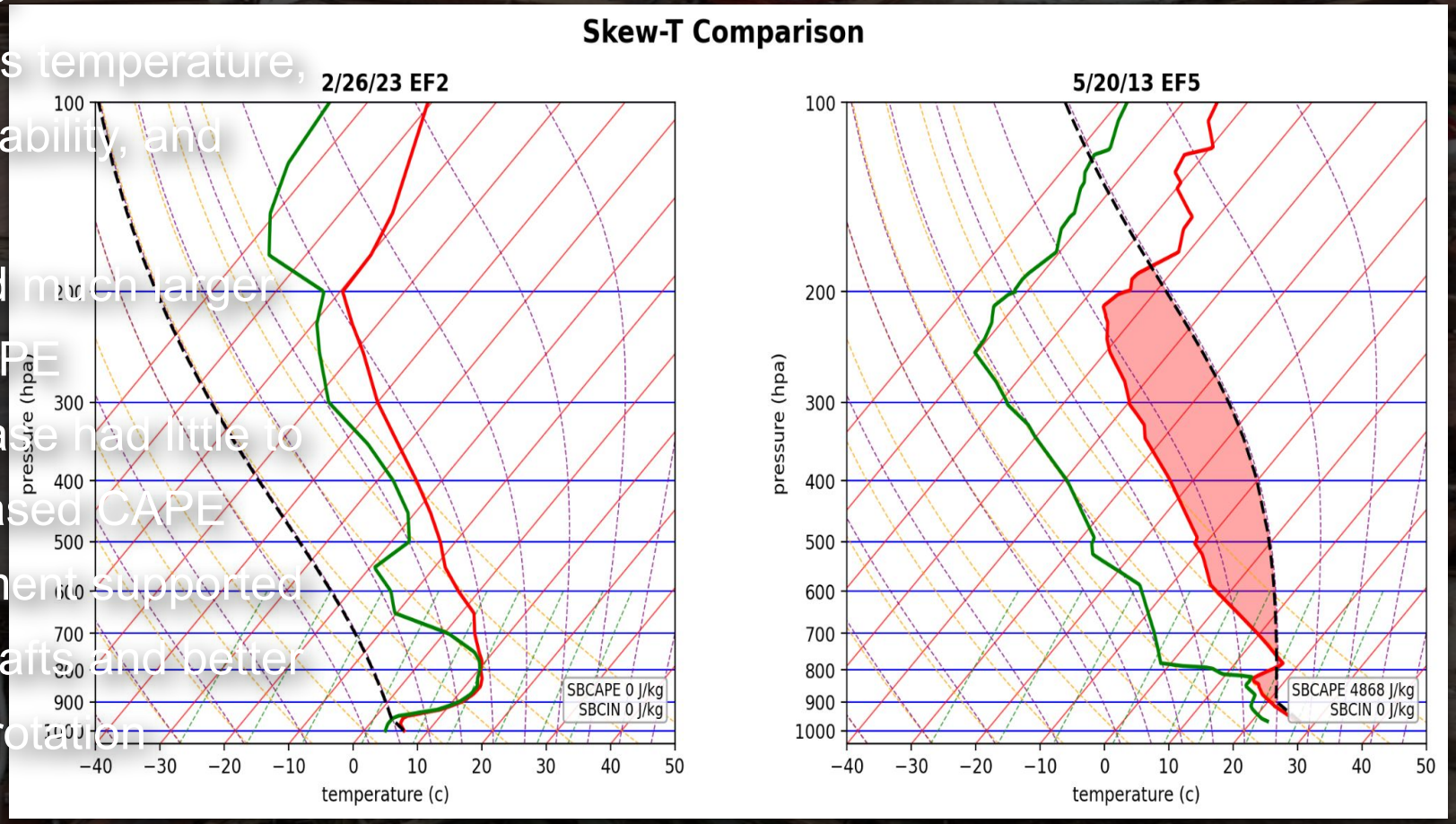
Wind-profile shape helped explain storm mode: classic supercell versus QLCS tornado environment.



ANALYSIS

- ★ Compares thermodynamic profiles for the QLCS EF2 and EF5 case
- ★ Skew-T shows temperature, moisture, instability, and inhibition
- ★ EF5 case had much larger usable SBCAPE
- ★ QLCS EF2 case had little to no surface-based CAPE
- ★ EF5 environment supported stronger updrafts and better stretching of rotation

The EF5 case had a much more explosive thermodynamic profile, while the QLCS case relied more on shear and forcing.



SUMMARY

- ★ Tornado intensity was not controlled by one parameter alone; the strongest events depended on ingredient overlap.
- ★ Violent tornado cases generally had strong instability, steep lapse rates, and favorable shear/SRH combinations.
- ★ May 10, 2010 had the strongest composite environment, even though the EF5 cases had higher CAPE.
- ★ The Feb. 26, 2023 QLCS case showed that tornadoes can occur with little SBCAPE when shear and forcing are strong.
- ★ CAPE and SRH were the most difficult parameters to calculate because they require accurate vertical profiles, parcel paths, storm motion, and layer integration.

The analysis showed that destructive tornado environments are produced by overlapping ingredients, but accurately calculating those ingredients especially CAPE and SRH was one of the biggest challenges. The main uncertainty came from representing the exact tornado environment using nearby soundings and reanalysis data.